

DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: **Computer Networks** | Course Code: **21CS52** | Credits: **4**

Exam Session: End Semester Exam 2024

Max Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. An organization is granted IP block 192.168.10.0/24. Design 4 subnets: Subnet A (60 hosts), Subnet B (30 hosts), Subnet C (20 hosts), Subnet D (10 hosts) using VLSM. List Subnet Address, Subnet Mask, Valid Host IP Range, and Broadcast Address for each. [8 Marks]

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Q2. Differentiate between Go-Back-N and Selective Repeat sliding window protocols. For a channel with bandwidth 1 Mbps, round trip time 40ms, and packet size 1000 bits, compute the link utilization for window size $N = 4$. [8 Marks]